<u>Point-by-point Response to Comments from five Reviewers</u>

First of all, we would like to express our sincere appreciation for five reviewers' in-depth critiques, which significantly helped us in preparing a revision of the paper. In the revised manuscript, we have tried our best to ***address all concerns from all reviewers***.

[Reviewer #1]

1. *There is a minor question, which is in the Step 2, "initial miRNA screening via Venn diagrams". I believe the authors should consider the over- and under- expression of the microRNAs, but it might be better to emphasize this.*

   Response: As suggested, we added in the subsection B under Section "Methods" that DEMs identified from GEO (Step 1) include both over-expressed and under-expressed miRNAs.

[Reviewer #2]

Major:

1. *The datasets they used are very heterogeneous. One compares primary and recurrent samples. The other two compare healthy versus disease tissue. All three datasets examine different tissue types. I understand that the researchers may want to look for miRNA that are observed across such heterogeneous conditions, but this application does not seem relevant to real-world biomarker applications. How do we know what the biomarkers ostensibly would predict?*

   Response: As explained in the subsection A under Section "Results and Discussion," three different datasets were chosen because they can well cover various aspects of GBM disease and its different stages.

2. *The authors used tool and methods that not reproducible at all. They used graphic user interface tools (GEO2R, Venny, OmniSearch). Although it is not necessarily inappropriate to use a GUI, the authors have not provided step-by-step instructions on their executed their analyses with these tools. The authors stated that "we conducted comprehensive semantic searches and analysis in the OmniSearch tool" and "Based on the comprehensive analysis of the federated knowledge returned in OmniSearch, we further filtered out most likely miRNAs as candidate biomarkers in GBM" and "We conducted semantic searches and analysis in OmniSearch for each of the ten miRNAs, and filtered out three most likely regulating miRNAs: hsa-miR-142-3p, hsa-let-7c, and hsa-miR-140-5p, which we believed are strongly potential biomarkers in GBM patients." But these are vague statements. It is impossible to fully understand their precise methodology. What caused them to believe that these particular miRNA had strong potential?*

   Response: All steps are clearly explained and detailed in the paper, along with screenshots to help users to better understand (and reproduce) results from our methodologies. For example, we stated in the subsection C under Section "Methods" that, "Based on the comprehensive analysis of the federated knowledge returned in OmniSearch, we further filtered out most likely miRNAs as candidate biomarkers in GBM." Then we detailed our analysis in the subsection C under Section "Results and Discussion" — "Our further filtering was based on the following observations…" along with Figures 4 to 7.

3. *Figure 1 is unhelpful. I could just browse to the GEO page to view information about this dataset. There is no reason to include a screenshot of it in the manuscript.*

   Response: Figure 1 is essential in the sense that it details Step 1 in our methods.

4. *Table 1 is also not helpful. I could easily obtain the same information on the GEO page. This table provides no new information.*

   Response: Similarly with Figure 1, Table 1 is also an essential part, which provides the readers with detailed information on samples from the GEO Series GSE32466.

5. *Table 2 summarizes the differential expression results. But the reader doesn't need all this information. They have no point of reference for t and B values, and I don't see the value in providing nominal p-values in addition to adjusted p-values.*

   Response: We explained in the paper that, t represents moderated t-statistic, and B represents B-statistic or log-odds that the miRNA is differentially expressed. Additional references were also added for both t-statistic ([30][31]) and B-statistic/log-odds ([32][33]). There is no nominal p-value provided in the GEO2R tool.

6. *I am confused as to why Table 2 contains control probes. In theory, these should not be differentially expressed. I am concerned that the authors found these to be differentially expressed. At a minimum, I don't think it makes sense to include them in the paper.*

   Response: Control probes can provide important clues in analyzing DEMs in GEO2R.

7. *How do we know that the 10 intersecting miRNA were not identified merely due to chance?*

   Response: As explained in the paper, the intersection among different sets of miRNAs was obtained through set theory using Venn diagrams.

8. *Figures 4-7 are do not provide much helpful information.*

   Response: The importance of Figures 4 to 7 was already explained above in #2. That is, to help users to better understand, and reproduce, our results.

9. *The authors failed to cite the paper that describes the creation of this dataset.*

   Response: As suggested, we added appropriate references ([27-29]).

Minor:

1. *The title is quite long and difficult to decipher. I would suggest making it more succinct.*

   Response: As suggested, we modified the title so that it is informative and straightforward.

2. *The abstract mentions examining three "expression profiles." However, I believe the authors*

meant that they had examined three data sets, each consisting of multiple profiles.

Response: As suggested, we modified the Abstract.

3. *Although Venn diagrams may be useful to visualize intersections among sets of DEGs, the underlying logic that they used for this is set theory. Besides, Venn diagrams are not very informative. I recommend the UpSetR package rather than Venn diagrams for examining intersections.*

   Response: As suggested, we included in Section “Conclusions” the discussion of utilizing the UpSetR package in future work.

4. *The text says Table 2 contains 250 DEMs, but there are only about 20 shown.*

   Response: As suggested, we modified the text and clearly stated that only the first 19 DEMs are shown in Table 2.

5. *Figure 3 is unnecessary.*

   Response: Figure 3 is necessary because it allows the readers to obtain the details of the intersection between pairwise DEM sets.

6. *Figures 5 and 7 are extremely low resolution.*

   Response: As suggested, we reproduced both figures with higher resolution.

[Reviewer #3]

Minor comments:

1. *Section II (A): how did you decide the two threshold values?*

   Response: We explained in the paper that these two threshold values were based on common practice in statistics.

2. *Fig 2 and Fig 3 must contain more explanation. It is hard to follow without a better explanation.*

   Response: As suggested, we added explanation to both figures as follows.

   Figure 2 visualizes – using the GEO Profile graphics – different expression levels of hsa-miR-142-3p, which is one of the 250 DEMs obtained from GSE32466. In Figure 2, it is clearly demonstrated that expression of hsa-miR-142-3p was significantly different between primary (left 12 bars) and recurrent (right 12 bars) GBM tissues.

   Figure 3 exhibits the initial screening results via the application of Venn diagrams in Venny 2.1, including intersections between pairwise GEO series, as well as common intersections among all three GEO series. There were 33 common DEMs between GSE32466 and GSE51322, 55 common DEMs between GSE32466 and GSE62381, 34 common DEMs between GSE51322 and GSE62381; and finally, there were 10 common DEMs among all three GEO series: hsa-miR-142-3p, hsa-miR-124, hsa-miR-140-5p, hsa-miR-939, hsa-miR-663, hsa-miR-503, hsa-miR-1281, hsa-let-7c, hsa-miR-15a, and hsa-let-7. These ten miRNAs are likely regulating miRNAs in GBM, subject for further semantic analysis.

3. *I recommend to avoid the use of acronyms in the abstract.*

   Response: As suggested, we removed all acronyms in Abstract.

[Reviewer #4]

1. *The paper is very well-written, covering important subject with innovative method.*

   Response: No edits are needed.

[Reviewer #5]

1. *Using Venn diagrams to filter out more likely miRNAs from multiple studies is too simple. A better, but still simple, approach would be to use Fisher's method.*

   Response: As suggested, we included in Section “Conclusions” the discussion of utilizing Fisher’s method in future work.